
MASTER SOLUTION — UNIT I: INTRODUCTION TO COMPUTER NETWORKS AND PHYSICAL LAYER		
Subject: Computer Networks	Branch: Computer Engineering	
Pattern: 2024	Sem: V	Total Questions: 15
Introduction to computer networks; uses of computer networks – business applications, home applications, mobile users; network hardware – PAN, LAN, MAN, WAN and internetworks; network software – protocol hierarchies, design issues for layers, connection-oriented and connection-less services, service primitives, relationship between services and protocols; reference models – OSI and TCP/IP models. Physical Layer: guided transmission media; wireless transmission; telephone system; narrowband and broadband communication systems		

Q1	Explain the basic building blocks and utility of computer networks. Describe business, home, and mobile applications of computer networks with suitable examples. [8M]
Ans.	A computer network is an interconnected collection of autonomous computing devices that are capable of exchanging information and sharing resources with one another using a common set of communication rules called protocols. The devices, referred to as nodes or hosts, are connected through transmission links such as copper cable, optical fibre or wireless radio channels, and the overall arrangement allows data, hardware and software resources to be used jointly rather than in isolation. The fundamental building blocks of a computer network can be understood in terms of the elements that must be present before any communication can take place. These elements work together so that a message generated at a source node reaches the intended destination node correctly and efficiently. • **End Devices (Nodes/Hosts):** These are the computers, servers, smartphones, printers and other terminal devices that originate or receive data. Every end device is assigned a unique address so that it can be identified uniquely on the network. • **Transmission Media:** This is the physical or wireless path through which signals travel between devices. It may be guided, such as twisted pair, coaxial cable and optical fibre, or unguided, such as radio waves, microwave and infrared. • **Networking Devices:** Hubs, switches, routers, bridges and gateways are intermediate devices that forward, filter or translate data as it moves from the source to the destination across one or more networks. • **Protocols:** A protocol is a formal set of rules that governs the format, timing, sequencing and error control of data exchanged between communicating entities. Without an agreed protocol, two devices cannot interpret each other's signals correctly.

- **Network Software:** Operating system components, network interface drivers and application-layer software together implement the protocol stack and present networking services to the end user in a usable form.

The utility of computer networks arises because they allow geographically distributed users and machines to share resources, communicate instantly and access common data. This sharing improves efficiency, reduces the cost of duplicating expensive resources, and increases the reliability of an organisation because data can be replicated across multiple machines. Based on who uses these benefits and how, network applications are broadly classified into business applications, home applications and mobile user applications.

- **Business Applications:** In a corporate environment, networks primarily target resource sharing and efficient communication among employees. A common example is a client-server model where several client machines access a centralised database server for order processing, payroll or inventory management, avoiding duplication of records at every desk.
- **Business-to-Business Commerce:** Organisations use networks to place purchase orders, verify invoices and track shipments electronically with suppliers, cutting down paperwork and turnaround time. Electronic Data Interchange (EDI) over a network is a classic example used by large manufacturing supply chains.
- **Business-to-Consumer Commerce:** Companies expose their catalogues and services over the network so that customers can browse, order and make payments online, as seen in e-commerce portals such as Amazon and Flipkart.
- **Collaborative Working:** Networks enable video conferencing, shared document editing and groupware applications so that geographically separated teams can work jointly on the same project in real time.
- **Home Applications:** At the household level, networks give individual users access to information, entertainment and communication. Access to remote information such as browsing news portals, digital libraries and government websites is one of the earliest and still dominant home uses.
- **Person-to-Person Communication:** Electronic mail, instant messaging platforms such as WhatsApp, and video calling applications such as Zoom allow family members and friends to stay connected irrespective of physical distance.
- **Interactive Entertainment:** Streaming of movies and music through services such as Netflix and Spotify, along with online multiplayer gaming, forms a major share of home network traffic today.
- **Electronic Commerce for Individuals:** Home users make use of online banking, bill payment and shopping applications, all of which depend on a reliable and secure network connection to a remote server.

- **Mobile User Applications:** Mobile users connect to networks through laptops, tablets and smartphones while moving from one location to another, so the network must support handoff between access points or cell towers without breaking the session.
- **Text and Multimedia Messaging:** Short Message Service, multimedia messaging and social networking applications such as Instagram and Facebook allow mobile users to exchange information instantly using cellular or Wi-Fi data.
- **Location Based Services:** Global Positioning System integrated with mobile networks allows navigation applications such as Google Maps to provide turn-by-turn directions and locate nearby services.
- **Mobile Commerce and Banking:** Digital wallets and Unified Payments Interface based applications let a mobile user complete a financial transaction instantly while travelling, relying entirely on the reliability of the underlying wireless network.

In summary, business, home and mobile applications together demonstrate that the true utility of a computer network is not confined to a single environment. The same underlying building blocks of nodes, media, devices and protocols are reused across all these domains, and only the scale, mobility requirement and traffic pattern change from one application category to another. This is precisely why a single organisation can simultaneously run a business database over its LAN, allow employees to check personal mail from home over the same Internet connection, and support field staff accessing the corporate system from mobile devices, all using the identical set of networking principles.

Q2	**Differentiate between LAN, MAN and WAN with respect to area covered, ownership, speed and technology used. [6M]**

Ans. Networks are commonly classified on the basis of the geographical area they span, because the physical distance directly affects the choice of transmission technology, the achievable data rate and the ownership pattern of the infrastructure. The three principal categories used for this classification are Local Area Network (LAN), Metropolitan Area Network (MAN) and Wide Area Network (WAN).

- **Local Area Network (LAN):** A LAN connects computers within a single building, floor or campus, typically spanning a few hundred metres to a few kilometres. It is owned privately by the organisation that uses it and offers very high data rates with low error rates because the transmission distance is short.
- **Metropolitan Area Network (MAN):** A MAN covers a larger geographical area such as an entire city, typically up to about 50 kilometres. It is often used to interconnect several LANs belonging to different branches of the same organisation and may be owned by a single company or leased from a service provider such as a cable television or telephone company.
- **Wide Area Network (WAN):** A WAN spans a country, continent or even the whole globe, as in the case of the Internet. Because it covers very large distances, it is generally owned